

1: Hello World RESTful Web Service

HelloController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;

@RestController

public class HelloController{

    private static final Logger LOGGER=LoggerFactory.getLogger(HelloController.class);

    @GetMapping("/hello")
    public String sayHello(){
        LOGGER.info("START");
        String message="Hello World!!";
        LOGGER.info("END");
        return message;
    }
}
```

Output

Request:

http://localhost:8083/hello

Response:

Hello World!!

Result: The REST service returns "**Hello World!!**" successfully.

2: REST – Country Web Service

CountryController.java

```
package com.cognizant.springlearn.controller;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.context.ApplicationContext;
import org.springframework.context.support.ClassPathXmlApplicationContext;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Country;

@RestController

public class CountryController{

    private static final Logger LOGGER=LoggerFactory.getLogger(CountryController.class);

    @RequestMapping("/country")
    public Country getCountryIndia(){
        LOGGER.info("START");
        ApplicationContext context=new ClassPathXmlApplicationContext("country.xml");
        Country country=context.getBean("country",Country.class);
        LOGGER.info("END");
        return country;
    }
}
```

country.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<beans xmlns="http://www.springframework.org/schema/beans"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.springframework.org/schema/beans
```

```
https://www.springframework.org/schema/beans/spring-beans.xsd">
```

```
<bean id="country" class="com.cognizant.springlearn.Country">
```

```
<property name="code" value="IN"/>
```

```
<property name="name" value="India"/>
```

```
</bean>
```

```
</beans>
```

Output

```
{  
  "code": "IN",  
  "name": "India"  
}
```

Result: The REST service loads the **Country** bean from the Spring XML configuration and returns it as JSON.

3: REST – Get Country Based on Country Code

CountryService.java

```
package com.cognizant.springlearn.service;
```

```
import java.util.ArrayList;
```

```
import org.springframework.context.ApplicationContext;
```

```
import org.springframework.context.support.ClassPathXmlApplicationContext;
```

```
import org.springframework.stereotype.Service;
```

```
import com.cognizant.springlearn.Country;
```

```
@Service
```

```
public class CountryService{
```

```
    public Country getCountry(String code){
```

```
ApplicationContext context=new ClassPathXmlApplicationContext("country.xml");
ArrayList<Country> countries=context.getBean("countryList",ArrayList.class);

for(Country country:countries){
    if(country.getCode().equalsIgnoreCase(code)){
        return country;
    }
}
return null;
}
}
```

CountryController.java

```
package com.cognizant.springlearn.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.PathVariable;
import org.springframework.web.bind.annotation.RestController;
import com.cognizant.springlearn.Country;
import com.cognizant.springlearn.service.CountryService;

@RestController

public class CountryController{

    @Autowired
    private CountryService countryService;

    @GetMapping("/country/{code}")
    public Country getCountry(@PathVariable String code){
```

```
        return countryService.getCountry(code);
    }
}
```

country.xml

```
<bean id="in" class="com.cognizant.springlearn.Country">
  <property name="code" value="IN"/>
  <property name="name" value="India"/>
</bean>
```

```
<bean id="us" class="com.cognizant.springlearn.Country">
  <property name="code" value="US"/>
  <property name="name" value="United States"/>
</bean>
```

```
<bean id="de" class="com.cognizant.springlearn.Country">
  <property name="code" value="DE"/>
  <property name="name" value="Germany"/>
</bean>
```

```
<bean id="jp" class="com.cognizant.springlearn.Country">
  <property name="code" value="JP"/>
  <property name="name" value="Japan"/>
</bean>
```

```
<bean id="countryList" class="java.util.ArrayList">
  <constructor-arg>
    <list>
      <ref bean="in"/>
      <ref bean="us"/>
```

```
<ref bean="de"/>
<ref bean="jp"/>
</list>
</constructor-arg>
</bean>
```

Output

Request:

<http://localhost:8083/country/in>

```
{
  "code": "IN",
  "name": "India"
}
```

Result: The REST service successfully retrieves a country based on the country code (case-insensitive) and returns the corresponding country details in JSON format.